Mahmoud Alyosify

NLP Engineer

New Cairo, Egypt · Open to relocation and remote · +20 114 555 7959 · mahmoud.alyosify@gmail.com
linkedin.com/in/mahmoudalyosify · github.com/MahmoudAlyosify · mahmoudalyosifysite.github.io

SUMMARY

NLP engineer working on **search, question answering, fine-tuning and evaluation** of language models. Fine-tuned Llama-3-8B on 159,826 records, built document question answering with hybrid search, and wrote an MSc thesis on shorter, complete LLM answers (47.6% fewer tokens). MSc in Artificial Intelligence, Queen's University, Canada (GPA 4.30 / 4.30).

SKILLS

NLP: Transformers, RAG, semantic and hybrid search, re-ranking, information extraction, text classification
Models: fine-tuning (LoRA, QLoRA), Llama, Mistral, Qwen
Tools: Hugging Face, PyTorch, LangChain, FAISS, NLTK
Evaluation: BERTScore, ROUGE, LLM-as-judge
Programming: Python, SQL

EXPERIENCE

AI Engineer Intern — Injaz Tech (Injaz Group), Cairo — Oct 2026 – Present

- Built question answering over company documents with hybrid search (BM25 + embeddings), **re-ranking** and query rewriting; top-1 retrieval accuracy rose from **75% to 94%**.
- Designed the evaluation: retrieval benchmarks (top-1, MRR), category-level scores and an LLM judge checked against a fixed metric (agreement 0.82).

Part-Time Machine Learning Instructor — National Telecommunication Institute (NTI), Egypt — Jul 2025 – Present

- Teach **90-hour machine learning bootcamps** and AI for Business modules in NTI's Summer Internships Program.

Systems Engineer (Military Service) — National Company for Roads Building & Development, Cairo — Feb 2024 – Mar 2025

- Supported the backend systems of Egypt's national toll network, used by **more than half of the country's vehicles**.

Online Instructor — Udemy & YouTube (Arabic courses), Remote — 2020 – Present

- Created Arabic courses on algorithms, discrete mathematics, probability and statistics: **10,000+ learners, 700K+ views**.

PROJECTS

MinimalLM — Master's thesis (Python, PyTorch, Hugging Face Transformers) — 2026

- A training-free method that makes open LLMs (Mistral-7B, Llama-3.1-8B, Qwen2.5-3B) give shorter answers that are still complete: **47.6% fewer tokens, 19–68% faster, no answer cut off** (0 of 990).

Horus-OSINT — fine-tuned LLM on AWS (Llama-3-8B, QLoRA, AWS) — 2026

- Fine-tuned **Meta-Llama-3-8B** on **159,826** global event and terrorism records (GTD, GDELT) and deployed it on **AWS** as an assistant that answers questions in plain language.

HORUS Sentinel — multi-agent intelligence platform (Python, LLM agents, knowledge graph)

- Turns continuously collected text into a knowledge graph of entities and relations, which a self-hosted LLM uses to write sourced reports.

EDUCATION

MSc in Artificial Intelligence — Queen's University, Kingston, Canada — Sep 2025 – Oct 2026

- GPA **4.30 / 4.30** · Completed October 2026 · Digilians Presidential Scholarship
- Thesis: MinimalLM — shorter LLM answers that stay complete (supervisor: Dr. Rahatara Ferdousi)

BSc in Computer & Information Science (Bioinformatics) — Assiut University, Egypt — Sep 2019 – Jul 2023

- GPA 3.53 / 4.00 (Very Good with Honours) · Graduation project graded A+

CERTIFICATIONS & AWARDS

IBM AI Developer Professional Certificate (IBM, 2026) · Deep Learning Specialization (DeepLearning.AI, 2026) · Machine Learning Specialization (DeepLearning.AI) · 1st Place, Smart Cities Hackathon 2022 (77 university teams)

LANGUAGES

Arabic (native) · English (professional working proficiency)